3.8 FLAIR & INVERSION RECOVERY SEQUENCES

SEEING WHAT T1 AND T2 CANNOT

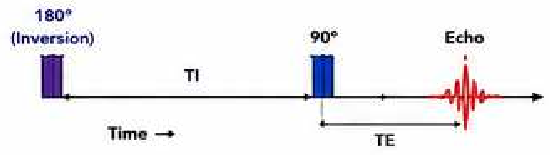
FLAIR (Fluid-Attenuated Inversion Recovery) is an inversion recovery sequence with long TI that nulls free fluid (CSF). Other IR sequences null different tissues for specialized applications.

KEY CONCEPT

- Inversion pulse (180°) → wait TI → 90° excitation → echo/train.
- TI (Inversion Time) determines which tissue is nulled.
- FLAIR = long TI to null CSF (long T1 fluid).
- Other IR sequences null fat, short T1 tissues, or fluids depending on TI.

↑ Increase / Benefit
↓ Decrease / Trade-off
— No / Minimal Effect

1 HOW INVERSION RECOVERY WORKS



- 180° pulse inverts the net magnetization (Mz into -Mz.
- Tissues recover toward +Mz at different rates (T1 dependent).
- Choose TI so a target tissue crosses through zero (Mz = 0).
- 90° pulse excites → nulled tissue produces little/no signal.
- TE controls T2 weighting of remaining signal.

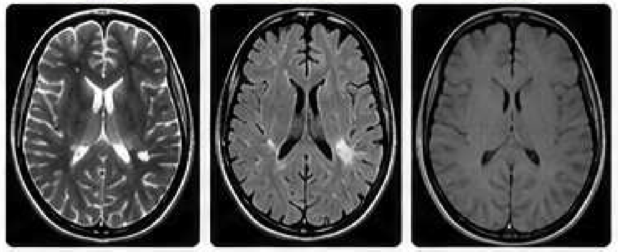
★ IR sequences are sensitive to B0 inhomogeneity and TI accuracy.

2 COMMON INVERSION RECOVERY VARIANTS

Sequence	Nulls / Suppresses	Typical TI (1.5T)	Main Use
FLAIR	CSF (free fluid)	2200–2800 ms	Brain lesions, edema, MS, subtle pathology
STIR	Fat	150–200 ms	MSK edema, bone marrow, inflammation
IR (Short TI)	Short T1 tissues (myocardium, blood)	150–300 ms	Cardiac viability, late enhancement
DIR (Double IR)	CSF and fat	Two TI's	Spine, cranial nerves, black-blood
PSIR	Short T1 (partial null)	Variable	Cardiac LGE with better SNR
MPRAGE (IR GRE)	Background tissues	700–1100 ms	3D T1 brain, anatomy

ⓘ TI values depend on field strength. At 3T, TI is generally longer (~10–20%).

3 IMAGE APPEARANCE COMPARISON (BRAIN – 1.5T)



T2-weighted (FSE)	FLAIR	T1-weighted (SE)
CSF: Very bright	CSF: Dark (nulled)	CSF: Dark
Edema: Bright	Edema: Bright	Edema: Dark
Gray Matter: Intermediate	Gray Matter: Intermediate	Gray Matter: Darker
White Matter: Dimmer	White Matter: Bright	White Matter: Bright
Lesions: Bright	Lesions: Very conspicuous	Lesions: Variable

ⓘ FLAIR suppresses CSF → improves lesion conspicuity near ventricles and sulci.

4 WHEN TO USE WHICH?

Clinical Situation / Goal	Best Choice	Why?
Periventricular / juxtacortical lesions (MS)	FLAIR	Suppressed CSF makes lesions stand out.
Subarachnoid hemorrhage	FLAIR	Blood bright; CSF dark → easy to see small SAH.
Edema / tumor / infection	FLAIR	Highlights edema and pathology.
Acute infarct (hyperacute)	FLAIR + DWI	DWI detects early; FLAIR confirms and defines extent.
Musculoskeletal edema (marrow/soft tissue)	STIR	Fat suppression uniform even with B0 inhomogeneity.
Spine imaging	STIR or FLAIR	STIR for edema; FLAIR for lesions near CSF.
Cardiac viability (scar)	IR (Short TI) or PSIR	Nulls normal myocardium; scar appears bright.
Fat suppression when B_0 is poor	STIR	STIR less sensitive to B_0 and center frequency.

5 SIGNAL INTENSITY GUIDE (GENERAL – 1.5T)

Tissue	FLAIR	T2	T1	STIR
CSF (free fluid)	Dark (nulled)	Very Bright	Dark	Dark
Edema / Water	Bright	Bright	Dark	Bright
Gray Matter	Intermediate	Intermediate	Darker	Intermediate
White Matter	Bright	Dimmer	Bright	Bright
Fat	Bright	Bright	Bright	Dark (nulled)
Blood (acute)	Variable	Bright	Dark	Variable
Protein (high)	Bright	Bright	Bright	Bright

★ Signal depends on TR, TE, TI, flip angle, and pathology.

6 INVERSION TIME (TI) SELECTION – BASICS

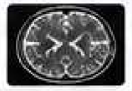
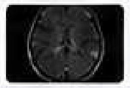
1. Know the target tissue to null (e.g., CSF, fat, myocardium).
2. Understand its T1 at your field strength.
3. Use vendor TI scout or equations to estimate TI (Mz = 0 crossing).
4. Fine-tune TI for uniform null across anatomy.
5. Re-check after major shimming changes or patient repositioning.

Typical TI Ranges (1.5T)

STIR (fat)	150 – 200 ms
IR (myocardium / short T1)	150 – 300 ms
FLAIR (CSF)	2200 – 2800 ms
MPRAGE (brain)	700 – 1100 ms
PSIR (cardiac LGE)	200 – 400 ms

ⓘ 3T: TI values are typically 10–20% longer than 1.5T.

7 COMMON ARTIFACTS & CAUSES

Artifact / Problem	Appearance	Common Causes	How to Reduce / Fix
Incomplete CSF null (FLAIR)		TI too short or too long B0 inhomogeneity Patient off-center	Adjust TI (scout) Improve shim / use volume shim Center FOV on anatomy
Patchy fat suppression (STIR if using fat sat instead)		Poor B0 homogeneity Incorrect center frequency Large anatomy	Use STIR instead of spec fat sat Improve shim / 3D shim Set correct center frequency
Global signal loss (IR sequences)		TI far from null point Wrong TI for anatomy Sequence parameters	Re-optimize TI Check sequence prescription
Ghosting / banding		Motion Flowing CSF Poor shimming	Improve immobilization Use flow compensation Re-shim

8 ENGINEER CHECKLIST (FLAIR / IR)

- ☑ Correct TI for target tissue and field strength?
- ☑ Good B0 homogeneity (shim test passed)?
- ☑ Is patient centered in the magnet?
- ☑ Coil element functioning (no dropouts)?
- ☑ No RF/gradient artifacts or spikes?
- ☑ TR / TE / TI consistent with protocol?
- ☑ SNR adequate? Consider NEX / bandwidth.
- ☑ Repeat TI scout if anatomy changed.

QUICK REFERENCE

Sequence	What it Nulls
FLAIR	CSF
STIR	Fat
IR (Short TI)	Short T1 (myocardium, blood)
DIR	CSF + Fat
PSIR	Normal Myocardium
MPRAGE	Background tissues

9 CLINICAL EXAMPLES (WHY FLAIR IS CRITICAL)

Multiple Sclerosis	Periventricular & juxtacortical lesions bright on FLAIR.
Small Subarachnoid Hemorrhage	Blood bright; CSF suppressed → easy detection.
Encephalitis / Infection	Edema and inflammation conspicuous.
Tumor / Mass	Peritumoral edema bright; helps define extent.
Hydrocephalus Evaluation	Ventricular size seen without CSF glare.

10 PARAMETERS THAT MATTER MOST

Parameter	Increase	Decrease	Effect on FLAIR
TI	More CSF null (higher TI)	Less CSF null (lower TI)	Controls null point. Too low = CSF bright. Too high = less lesion contrast.
TR	More SNR	Less SNR	Higher TR improves SNR and CSF null.
TE	More T2 contrast	Less T2 contrast	Increase TE slightly for edema contrast, watch SNR.
Bandwidth	Less distortion	More SNR	Lower BW increases SNR but more chemical shift.
NEX	More SNR	Less scan time	Use higher NEX for better image quality.

11 PRACTICAL TIPS

- Always run a TI scout for FLAIR on new anatomy or after moving patient.
- Use 3D FLAIR for better CSF suppression and isotropic resolution.
- For poor shim areas, STIR is more reliable than spectral fat sat.
- Keep patient still—motion is very visible on long TR sequences.
- Review images on all planes; small lesions hide on single slice.
- Compare with T1, T2, DWI for complete clinical picture.

REMEMBER

FLAIR = long TI → nulls CSF.

- STIR = short TI → nulls fat.
- IR is a powerful tool—TI is everything.
- Optimize TI, TR, and shim to get the right image.

KEY TAKEAWAY

IR sequences reveal pathology hidden by fluid or fat.

Master TI selection and B_0 homogeneity to produce clear, diagnostic images.

★ RIGHT INFORMATION. RIGHT BALANCE. RIGHT IMAGE. Understand the sequence. Recognize the artifacts. Support the technologist. Deliver the value.